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Named Entity Recognition in German-language Telegram channels

Bachelor Thesis

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Faculty 4

by

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Abstract

This thesis implements Named Entity Recognition (NER) in German-language text data disseminated via the messenger service Telegram, with a focus on the mobilization for protests against measures to counteract the COVID-19 in Germany.

This work explores the potentials and limitations of two existing German-language models for NER, namely *spacy* and *hugging face*, for the recognition of the entity types PERSON, LOCATION, and ORGANIZATION.

Furthermore, it makes use of the *ctparse* module to extract text parts containing DATE and TIME information.

This thesis contributes to the task of German-language NER in general by applying it to a specific domain and data type, namely text data from messenger services. Furthermore, its aim is to provide a low-resource tool for both researchers and civil society for monitoring and analyzing large amount of text data from social media/messengers.

Dedication

Statutory Declaration

I herewith formally declare that I have written the submitted thesis independently. I did not use any outside support except for the quoted literature and other sources mentioned in the paper.

I clearly marked and separately listed all of the literature and all of the other sources which I employed when producing this academic work, either literally or in content.

I am aware that the violation of this regulation will lead to the failure of the thesis.

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Chapter 1

Introduction

1.1 Background

The outbreak of the COVID-19 pandemic was followed by an increasing spread of misinformation and conspiracy theories. Those narratives evolved not only around the virus itself, but also about the intentions of those responsible for putting into effect measures for its containment.

Those measures included the mandatory use of face masks, rapid testing, and vaccinations, at times functioning as admission requirements in the workplace or publicly accessible spaces such as restaurants and theatres. Furthermore, unprecedented restrictions were imposed on public life through several general or partial lock-downs of social and economic life, as well as temporary bans of gathering in public. These restrictions also heavily affected the possibilities to hold political demonstrations.

From the beginning on, these measures were accompanied by critique and protests from various political spectres. In Germany, the first noticeable protests against COVID-19 measures took place at the so-called ‘Hygienedemonstrationen’ (hygiene demonstrations) in front of the ‘Volksbühne’ theatre in Berlin. Over time, the so-called ‘Querdenken’ (lateral thinking) movement became the most prominent movement evolving around protests

against the COVID-19 containment measures. The movement reached its preliminary climax with several large demonstrations taking place in Berlin, directed against legislative changes regarding the containment of the pandemic passed by the German parliament. In the context of these demonstrations, violent clashes occurred between police and protesters, which often refused to act according to the official restrictions for the demonstration such as wearing a face mask. Furthermore, there were several incidents in which journalists were violently attacked by protesters, accusing them to be part of the ‘Lügenpresse’ (lying press), and two incidents in which the seat of the German parliament, the Reichstag, was targeted by protesters: On August 29th 2020, around 500 protesters, among them several prominent actors from the German far-right movement, broke the barriers in front of the Reichstag in Berlin, seat of the German parliament, and occupied the stairs of the building, waving several ‘Reichskriegsflaggen’. On November 18th 2020, another large demonstration organized by the ‘Querdenken’ movement took place; once again accompanied by heavy clashes between protesters and police officers. On the same day, a small group of right-wing social media activists was invited as visitors to the Reichstag by the German far-right party ‘Alternative für Deutschland’ (AfD). The group approached and harassed politicians from other parties, filming and live-streaming the events, an incident which was perceived as a caesura and heavily criticized by the other parties.

On April 21st 2021, the last large authorized demonstration of the ‘Querdenken’ movement took place in Berlin, comparing a new law for the containment of the pandemic with the ‘Ermächtigungsgesetz’ put into effect in 1933 which ultimately paved the way for the Nazi dictatorship in Germany. In response to several violations of official restrictions by demonstration participants, police decided to disperse the demonstration. Several protesters subsequently moved to other areas nearby. Clashes between protesters and police continued to occur throughout the rest of the day.

The subsequent demonstrations announced by the ‘Querdenken’ move-

ment to take place in Berlin were prohibited by authorities, arguing with the experience at prior demonstrations that participants would act against the official restrictions. The ban was confirmed by Berlin's administrative court after the organizers of the protest had filed a suit against the initial prohibition. In spite of the ban, several hundred protesters gathered in different spots in Berlin e.g. on the 21st, 28th and 29th of August 2021.

The mentioned bans of large demonstrations of the 'Querdenken' movement as well as the general restrictions for demonstrations and public gatherings, limiting the number of participants to a maximum of twenty persons, decreased the momentum of the large, centralized protests focussing on the German legislative located in Berlin. However, the protests did not stop as a result, but rather changed their character: They became more decentralized and a shift to more informal forms of gatherings and protests could be observed: In several German cities and towns, people gathered for 'Mahnwachen' or 'Spaziergänge' (walks) to protest against the government's COVID-19 politics. Since going for a walk individually was a legal outdoor activity even under the strictest lock-down regulations, this type of action was chosen in order to circumvent the restrictions imposed on demonstrations. In effect, these walks ('Spaziergänge') often resulted in large gatherings of people moving through the streets together, often holding signs or chanting slogans against German politicians or the COVID-19 containment measures in general.

However, the calls to gather for walks in the public became more targeted and mobilized people to gather in front of the private homes of politicians, mostly those in charge for the COVID-19 measures on federal state level.

In December 2021, around thirty members of the far-right group 'Freie Sachsen' (Free Saxonians) gathered in front of the home of Saxony's Minister of Health, Petra Köpping, carrying torchlights. The incident received a lot of public attention, and was criticized in the media as well as by German politicians as attempts to intimidate politicians under the pretext of the

protesters right to ‘freedom of speech’.

Earlier that year, the same group had gathered in front of Saxony’s prime minister, Michael Kretschmer, who is one of the most prominent targets of the radical movement against the COVID-19 containment measures. In December 2021, an investigative report by German journalists revealed that members of the Telegram chat group ‘Dresden Offlinevernetzung’ had talked about concrete preparations to assassinate Kretschmer. A few months later, by mid-April 2022, police foiled a plot to kidnap Germany’s Minister of Health Karl Lauterbach and to sabotage power facilities in order to create chaos which would ultimately allow to overthrow the government.

The messenger service Telegram played a key role in the mobilization of the movement. The service allows the creation of large groups and channels to disseminate information quickly to a large number of users. Unlike Facebook or Google, which hand over data to authorities at their request, Telegram does not cooperate with authorities.

This ensures a high degree of anonymity for its users which has proven beneficial for protest movements in authoritarian countries, This anonymity poses a barrier to political repression and criminal prosecution. This makes the service attractive not only for social movements, but also for criminal and terrorist activities, and also facilitates the spread of hate-speech and calls for violence like those mentioned above.

The rise of conspiracy theories in the context of the COVID-19 pandemic has thus not only intensified hate-speech against and the discrimination of certain communities in the form of antisemitism and racism - this process was accompanied by an increasingly toxic and potentially violent climate for politicians in Germany. This makes it once more clear that the digital sphere of social networks and messengers cannot be grasped without also considering its entanglement with and effects for the physical sphere of society. Both online and offline contexts influence and often reinforce each other, as is the case with the mobilization against the COVID-19 containment measures.

The processes illustrated above also point out another important aspect: Even though political activists from Germany's extremist far-right had been involved in the protests from the start, and non-governmental organizations had warned about the antidemocratic and violent potential of parts of the movement, the dangerous potential of this movement seems to have been underestimated by German authorities for a rather long time. Only after journalists had revealed the plans to murder Michael Kretschmer, and after several gatherings in front of private homes of politicians had already taken place, the issue seems to become more of a serious concern.

Thus the process points out the relevance of investigative work by critical journalists and non-governmental organizations to monitor the mobilization and networking strategies of the far-right spectre in Germany. However, while these actors often have to deal with large amount of data, their investigation often relies on manually conducted research and data evaluation. The potentials of Data Science methods, especially Machine Learning-based approaches for Natural Language Processing (NLP), are often not accessible for these actors, making it hard for them to process large amounts of data from the often very dynamic contexts of informal political networks.

- racism: especially Asian community made responsible for spreading the virus
- antisemitism: Jews identified as culprits behind the pandemic, using it as a means to obtain world dominance
- politicians: complicit with those who benefit from pandemic; following a secret plan; restriction of civil rights
- freedom of speech, criticizing the government are valuable and essential components of a democratic society
- but: often hateful tendencies and developments, both online and offline

- entanglement of digital and real world
- hate does not stay in the internet, also has real effects
- spread of misinformation, conspiracy theories: social media and messengers
- not only information, also mobilization

1.2 Objective

This thesis aims at exploring the potentials of a particular NLP technology, namely Named Entity Recognition (NER) for the investigation of the mobilization for protests against COVID-19 measures in a German-language Telegram channel. The overarching goal is to explore if a component for Named Entity Recognition could be a beneficial contribution to the Open Source project ‘Teledash’, to make NER available as a tool for journalists, NGOs, and researchers investigating the COVID-19 related protest movements and networks in Germany.

- low-resource approach of NER in German-language Telegram messages
- extraction of entities particularly relevant for mobilization / monitoring of potential targeting: PERSON, ORGANIZATION; DATE; TIME; LOCATION; ACTION as custom type
- explore potentials and limitations of existing models
- explore potentials of translation approach
- provide a low-resource tool for NGOs and researchers make available as API; Project teledash

1.3 Thesis Structure

- Theory
- Implementation
- Conclusion and Discussion

Chapter 2

Theory

2.1 Natural Language Processing

2.1.1 Definition

2.1.2 Historical evolvement

2.1.3 Challenges for NER

2.2 NER approaches

2.2.1 Rule-based

2.2.2 Machine-learning based

2.2.3 Neural Networks

2.2.4 Summarization

Chapter 3

Implementation

3.1 Data

3.1.1 Existing German-language datasets for NER

3.1.2 Domain-specific data

Telegram channel ‘Demotermine’: 52.900 subscribers by the beginning of June 2022,

long channel name: ‘Demo-Termine Kontakte, Gleichgesinnte treffen - Demokalender, Spaziergänge, Mahnwachen, Meditation *Rücktritt Bundesregierung!’

channel-info: “ Wir Demokraten wollen miteinander in Kontakt bleiben @DemotermineChat und uns im Bedarfsfall möglichst schnell austauschen, Beweise in Wort, Bild und Ton sichern, damit sie nicht unterschlagen werden können @LivestreamsFuerDich! @Terminverteiler_{Bot}’

TODO insert histplot to show development of message numbers

date range: 2018-10-05 bis 2021-12-19

total number of messages (wo duplicates): 27.627 messages

German-language: 23.911

Sample size: xx out of 23.911 confidence level 95

text length: min 100 (lower quartile), max 350 (upper quartile)

-i remaining:

3.2 Models

- spacy - hugging face

3.3 Pipeline

3.4 Evaluation metrics

3.4.1 Accuracy

3.4.2 Performance

3.5 Translation approach?

3.6 Summarization

Chapter 4

Results

4.1 Spacy

4.2 Hugging Face

4.3 Translation approach

4.4 Summarization

Chapter 5

Conclusion

5.1 Summarization

5.1.1 subsection

5.2 Discussion

Appendix

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List of Acronyms